

1. Create a console application for various ATM transactions. the user has to select an option from the options displayed on the screen. The options are related to withdraw the money, deposit the money, check the balance, and exit.

using System;

namespace AtmMachine

{

class Program

{

static void Main(string[] args)

{

//declare and initialize balance, withdraw, and deposit
int balance = 100000, withdraw, deposit;

//create scanner class object to get choice of user

while (true)

{

Console.WriteLine("Automated Teller Machine");
Console.WriteLine("Choose 1 for Withdraw");
Console.WriteLine("Choose 2 for Deposit");
Console.WriteLine("Choose 3 for Check Balance");
Console.WriteLine("Choose 4 for EXIT");
Console.WriteLine("Choose the operation you want to perform:");

//get choice from user

int choice = Convert.ToInt32(Console.ReadLine());

switch (choice)

{

case 1:

Console.WriteLine("Enter money to be withdrawn:");

//get the withdrawl money from user

withdraw = Convert.ToInt32(Console.ReadLine());

//check whether the balance is greater than or equal to the

withdrawal amount

if (balance >= withdraw)

{

//remove the withdrawl amount from the total balance

balance = balance - withdraw;

Console.WriteLine("Please collect your money");

}

else

{

//show custom error message

Console.WriteLine("Insufficient Balance");

}

Console.WriteLine("");

break;

case 2:

Console.WriteLine("Enter money to be deposited:");

//get deposit amount from te user

deposit = Convert.ToInt32(Console.ReadLine());

//add the deposit amount to the total balance

balance = balance + deposit;

```

        Console.WriteLine("Your Money has been successfully
deposited");

        Console.WriteLine("");
        break;

    case 3:
        //displaying the total balance of the user
        Console.WriteLine("Balance : " + balance);
        Console.WriteLine("");
        break;

    case 4:
        //exit from the menu
        return;
    }
}
}
}
}
}

```

2. Create a desktop application to design a calculator that could perform addition, subtraction, multiplication and division.

```

using System;
using System.Windows.Forms;

namespace Calculator
{
    public partial class Form1 : Form
    {
        int x, y, z;
        void Calculate()
        {
            x = Convert.ToInt32(textBox1.Text);
            y = Convert.ToInt32(textBox2.Text);

        }
        public Form1()
        {
            InitializeComponent();
        }

        private void button3_Click(object sender, EventArgs e)
        {
            Calculate();
            z = x * y;
            textBox3.Text = Convert.ToString(z);
        }

        private void button4_Click(object sender, EventArgs e)
        {
            Calculate();
            try
            {
                z = x / y;
                textBox3.Text = Convert.ToString(z);
            }
            catch (Exception ex)
            {
                MessageBox.Show(Convert.ToString(ex));
            }
        }
    }
}

```

```

private void button1_Click(object sender, EventArgs e)
{
    Calculate();

    z = x + y;
    textBox3.Text = Convert.ToString(z);

}

private void button2_Click(object sender, EventArgs e)
{
    Calculate();
    z = x - y;
    textBox3.Text = Convert.ToString(z);
}
}

```

3. Create a console application to check whether the entered number is a spy?

Spy Number

A positive integer is called a spy number if the **sum** and **product** of its digits are equal. In other words, a number whose sum and product of all digits are equal is called a **spy number**.

```

using System;

namespace SpyNumber
{
    class Program
    {
        static void Main(string[] args)
        {
            int num, product = 1, sum = 0, lastdigit;
            // create object of scanner
            Console.WriteLine("Enter the number to check: ");
            //reads an integer from the user and stores it in the variable num
            num = Convert.ToInt32(Console.ReadLine());
            //executes untill the condition becomes false
            while (num > 0)
            {
                //finds the last digit of the number
                lastdigit = num % 10;
                //adds last digit to the variable sum
                sum = sum + lastdigit;
                //calculates the product
                product = product * lastdigit;
                //removes the last digit from the given number
                num = num / 10;
            }
            //compares the sum and product
            if (sum == product)
                //prints if the above condition returns true
                Console.WriteLine("The given number is a spy number.");
            else
                //prints if the above condition returns false
                Console.WriteLine("The given number is not a spy number.");
        }
    }
}

```

```

    }
}

```

4. Create a console application **to remove the duplicate entry in an array.**

using System;

```

namespace RemoveDuplicateinArray
{
    class Program
    {
        public static int removeDuplicateElements(int[] arr, int n)
        {
            if (n == 0 || n == 1)
            {
                return n;
            }
            int[] temp = new int[n];
            int j = 0;

            for (int i = 0; i < n - 1; i++)
            {
                if (arr[i] != arr[i + 1])
                {
                    temp[j++] = arr[i];
                }
            }
            temp[j++] = arr[n - 1];
            // Changing original array
            for (int i = 0; i < j; i++)
            {
                arr[i] = temp[i];
            }
            return j;
        }

        static void Main(string[] args)
        {
            int[] arr = { 10, 20, 20, 30, 30, 40, 50, 50 };
            int length = arr.Length;
            length = removeDuplicateElements(arr, length);
            //printing array elements
            for (int i = 0; i < length; i++)
                Console.WriteLine(arr[i] + " ");
        }
    }
}

```

5. **Create a console application to determine whether two strings are the anagram or not?**
using System;

```

namespace Anagram
{
    class Program
    {
        static void Main(string[] args)
        {

```

```

string str1 = "silent";
string str2 = "listen";
char[] ch1 = str1.ToLower().ToCharArray();
char[] ch2 = str2.ToLower().ToCharArray();
Array.Sort(ch1);
Array.Sort(ch2);
string val1 = new string(ch1);
string val2 = new string(ch2);

if (val1 == val2)
{
    Console.WriteLine("Both the strings are Anagrams");
}
else
{
    Console.WriteLine("Both the strings are not Anagrams");
}
}
}
}

```

6. Create a console application to print the elements of an array in Reverse order.
using System;

```

namespace ReverseArray
{
    class Program
    {
        static void Main(string[] args)
        {
            //Initialize array
            int[] arr = new int[] { 1, 2, 3, 4, 5 };
            Console.WriteLine("Original array: ");
            for (int i = 0; i < arr.Length; i++)
            {
                Console.WriteLine(arr[i] + " ");
            }
            Console.WriteLine();
            Console.WriteLine("Array in reverse order: ");
            //Loop through the array in reverse order
            for (int i = arr.Length - 1; i >= 0; i--)
            {
                Console.WriteLine(arr[i] + " ");
            }
        }
    }
}

```

7. Write a C# program to calculate the interest payable by the user. User must have following options:
- To select his bank among SBI, ICICI & AXIS,
 - To input Principal Amount,
 - To input year of loan

Given:

Bank	Rate of interest
SBI	8
ICICI	7
AXIS	9

using System;

```
namespace RateofInterest
{
    class Bank
    {
        int GetRateOfInterest()
        { return 0; }
    }
    //Creating child classes.
    class SBI : Bank
    {
        public int GetRateOfInterest()

        { return 8; }
    }

    class ICICI : Bank
    {
        public int GetRateOfInterest()

        { return 7; }
    }
    class AXIS : Bank
    {
        public int GetRateOfInterest()

        { return 9; }
    }

    class Program
    {
        static void Main(string[] args)
        {
            int num, p = 0, r = 0, t = 0;
            double interest;
```

```

SBI s = new SBI();
ICICI i = new ICICI();
AXIS a = new AXIS();

Console.WriteLine("Select your Bank");
Console.WriteLine("enter 1 for SBI");
Console.WriteLine("enter 2 for ICICI");
Console.WriteLine("enter 3 for AXIS");

num = Convert.ToInt32(Console.ReadLine());
if (num == 1)
{
    Console.WriteLine("SBI Rate of Interest: " +
s.GetRateOfInterest());

    Console.WriteLine("Enter Principal");
    p = Convert.ToInt32(Console.ReadLine());
    r = s.GetRateOfInterest();
    Console.WriteLine("Enter time period");
    t = Convert.ToInt32(Console.ReadLine());
}
else if (num == 2)
{
    Console.WriteLine("ICICI Rate of Interest: " +
i.GetRateOfInterest());

    Console.WriteLine("Enter Principal");
    p = Convert.ToInt32(Console.ReadLine());
    r = i.GetRateOfInterest();
    Console.WriteLine("Enter time period");
    t = Convert.ToInt32(Console.ReadLine());
}
else if (num == 3)
{
    Console.WriteLine("AXIS Rate of Interest: " +
a.GetRateOfInterest());

    Console.WriteLine("Enter Principal");
    p = Convert.ToInt32(Console.ReadLine());
    r = a.GetRateOfInterest();
    Console.WriteLine("Enter time period");
    t = Convert.ToInt32(Console.ReadLine());
}
else
{
    Console.WriteLine("Wrong Choice entered");
}
interest = (p * r * t) / 100;
Console.WriteLine("Interest Payable=" + interest);
}

```

```
    }  
}
```

8. The straight line method of computing the yearly depreciation of the value of an item is given by:

$\text{Depreciation} = (\text{Purchase} - \text{salvage value}) / \text{years of service}$

write a program to determine the salvage value of an item when the purchase price, years of service and the annual depreciation are entered by the user.

using System;

namespace SalvagePrg

{

class Program

{

double dep, purchase, salv;

int year;

double salvage(double d, double p, int y)

{

dep = d;

purchase = p;

year = y;

salv = dep * year - purchase;

return (salv);

}

static void Main(string[] args)

{

double result, d, p;

int y;

Console.WriteLine("Enter depretiation,principle and year respectively");

d = Convert.ToDouble(Console.ReadLine());

p = Convert.ToDouble(Console.ReadLine());


```

        y = Convert.ToInt32(Console.ReadLine());

        Program obj = new Program();

        result = obj.salvage(d, p, y);

        Console.WriteLine(result);

    }

}
}

```

9. The total distance travelled by a vehicle in t seconds is given by:

Distance = $ut + (at^2)/2$ where u is the initial velocity (meter per second), a is the acceleration (meter per second square). write a program to evaluate the distance travelled at regular intervals of time, given the values of u & a. The program should provide the flexibility to the user to select his own time intervals and repeat the calculations for different values of u & a.

using System;

```

namespace DistanceTravelled
{
    class Program
    {
        float u, t, a, s;

        float distance(float u1, float t1, float a1)
        {
            u = u1;
            t = t1;
            a = a1;

            s = u * t + (a * t * t) / 2;

            return (s);
        }

        static void Main(string[] args)
    }
}

```

```

{
    float result, u1, t1, a1;
    int choice;

    do
    {
        Console.WriteLine("Enter the values of u,t and a respectively");
        u1 = Convert.ToSingle(Console.ReadLine()); //converting in float
        t1 = Convert.ToSingle(Console.ReadLine());
        a1 = Convert.ToSingle(Console.ReadLine());

        Program obj = new Program();

        result = obj.distance(u1, t1, a1);
        Console.WriteLine(result);
        choice = 0;
        Console.WriteLine("If you want to continue press 1 else 0 ");
        choice = Convert.ToInt32(Console.ReadLine());
        if (choice != 1)
        {
            Console.WriteLine("Thanks");
            break;
        }
    } while (choice == 1);
}
}
}

```

10. For a certain electrical circuit with an inductance L and resistance R , the damped natural frequency is given by:

Frequency =

$$\sqrt{\frac{1}{LC} - \frac{R^2}{4C^2}}$$

It is desired to study the variation of this frequency with C (capacitance). Write a program to calculate the frequency for different values of C starting from point 0.01 to 0.1 in steps of 0.01.

```
using System;

namespace DampedFrequency
{
    class Program
    {
        private int l, c, r;
        float frequency;
        Program(int l, int c, int r)
        {
            this.l = l;
            this.c = c;
            this.r = r;
        }

        public float cal()
        {
            frequency = (float)Math.Sqrt((1.0 / (l * (float)c)) - ((float)r * r /
4.0 * c * c));
            return (frequency);
        }

        static void Main(string[] args)
        {
            float fq;
            int l, c, r;
            Console.WriteLine("Enter the values of l,c and r");
            l = Convert.ToInt32(Console.ReadLine());
            c = Convert.ToInt32(Console.ReadLine());
            r = Convert.ToInt32(Console.ReadLine());
            Program obj = new Program(l,c,r);
            try
            {
                fq = obj.cal();
                Console.WriteLine(Convert.ToString(fq));
            }
            catch(Exception e)
            { Console.WriteLine(e); }
        }
    }
}
```

11. Write a C# program to establish a connection using SqlConnection and display the status.

```
using System;
using System.Data.SqlClient;
```

```
class Program
{
```

```

static void Main()
{
    string connString = @"Data Source=YOUR_SERVER_NAME; Initial
Catalog=YOUR_DATABASE; Integrated Security=True";

    SqlConnection conn = new SqlConnection(connString);

    try
    {
        conn.Open();
        Console.WriteLine("Connection State: " + conn.State);
        Console.WriteLine("Connection Successfully Established!");
    }
    catch (Exception ex)
    {
        Console.WriteLine("Error: " + ex.Message);
    }
    finally
    {
        conn.Close();
        Console.WriteLine("Connection Closed. Current State: " + conn.State);
    }

    Console.ReadLine();
}
}

```

-
12. Write a C# program to Display Records Using SqlDataReader assuming table: Student(RollNo INT, Name VARCHAR(50), Marks INT).

```

using System;
using System.Data.SqlClient;

class Program
{
    static void Main()
    {
        string connString = @"Data Source=YOUR_SERVER;Initial Catalog=YOUR_DATABASE;Integrated
Security=True";

        using (SqlConnection conn = new SqlConnection(connString))
        {
            conn.Open();

            SqlCommand cmd = new SqlCommand("SELECT RollNo, Name, Marks FROM Student", conn);

            SqlDataReader dr = cmd.ExecuteReader();

            Console.WriteLine("RollNo\tName\tMarks");
            Console.WriteLine("-----");

            while (dr.Read())
            {

```

```

        Console.WriteLine(dr["RollNo"] + "\t" + dr["Name"] + "\t" + dr["Marks"]);
    }
}

Console.ReadLine();
}
}

```

13. Write a C# program to

- a. Load data into DataSet
- b. Modify records in DataSet
- c. Update database

```

using System;
using System.Data;
using System.Data.SqlClient;

class Program
{
    static void Main()
    {
        string connString = @"Data Source=YOUR_SERVER;Initial Catalog=YOUR_DATABASE;Integrated
Security=True";

        SqlConnection conn = new SqlConnection(connString);
        SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM Student", conn);

        SqlCommandBuilder cb = new SqlCommandBuilder(da);

        DataSet ds = new DataSet();
        da.Fill(ds, "Student");

        Console.WriteLine("Original Marks of RollNo=1: " + ds.Tables["Student"].Rows[0]["Marks"]);

        // Modify data inside DataSet
        ds.Tables["Student"].Rows[0]["Marks"] = 99;

        // Update database
        da.Update(ds, "Student");

        Console.WriteLine("Database Updated!");
        Console.ReadLine();
    }
}

```

14. Write a C# program to create a WinForm and do Insert, Update, Delete Using SqlCommand, assuming table: Student(RollNo INT, Name VARCHAR(50), Marks INT)

```

using System;
using System.Data.SqlClient;

class Program
{
    static void Main()
    {

```

```

string connString = @"Data Source=YOUR_SERVER;Initial Catalog=YOUR_DATABASE;Integrated
Security=True";

SqlConnection conn = new SqlConnection(connString);
conn.Open();

// INSERT
SqlCommand insertCmd = new SqlCommand(
    "INSERT INTO Student VALUES (4, 'Amit', 88)", conn);
insertCmd.ExecuteNonQuery();
Console.WriteLine("Record Inserted!");

// UPDATE
SqlCommand updateCmd = new SqlCommand(
    "UPDATE Student SET Marks=95 WHERE RollNo=4", conn);
updateCmd.ExecuteNonQuery();
Console.WriteLine("Record Updated!");

// DELETE
SqlCommand deleteCmd = new SqlCommand(
    "DELETE FROM Student WHERE RollNo=4", conn);
deleteCmd.ExecuteNonQuery();
Console.WriteLine("Record Deleted!");

conn.Close();
Console.ReadLine();
}
}

```

15. Write a C# program to Create a WinForms app using:

a. `SqlDataSource` for automatic binding

Bind it to:

b. `GridView` / `DropDownList`

and display database records.

```

using System;
using System.Data;
using System.Windows.Forms;
using System.Data.SqlClient;

public partial class Form1 : Form
{
    public Form1()
    {
        InitializeComponent();
        LoadData();
    }

    void LoadData()
    {
        string connString = @"Data Source=YOUR_SERVER;Initial Catalog=YOUR_DATABASE;Integrated
Security=True";

```

```
SqlDataAdapter da = new SqlDataAdapter("SELECT * FROM Student", connString);  
DataTable dt = new DataTable();  
da.Fill(dt);  
  
dataGridView1.DataSource = dt;  
}  
}
```